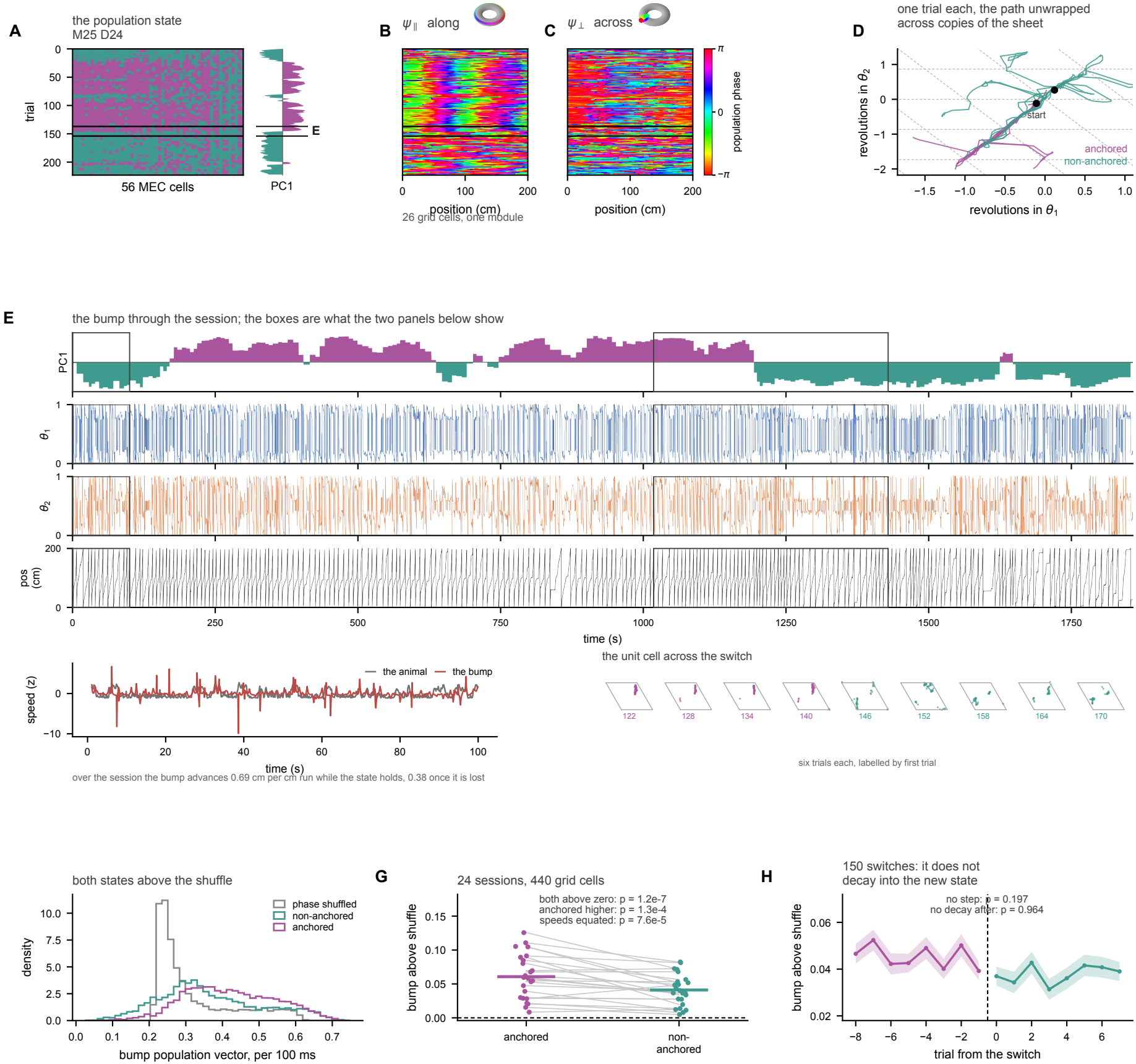

A
the population state
M25 D24
trial
56 MEC cells
PC1
E
B
$\psi_\parallel$ along
position (cm)
26 grid cells, one module
C
$\psi_\perp$ across
position (cm)
population phase
D
one trial each, the path unwrapped
across copies of the sheet
revolutions in θ_2
revolutions in θ_1
start
anchored
non-anchored
E
the bump through the session; the boxes are what the two panels below show
PC1
θ_1
θ_2
pos (cm)
time (s)
the animal
the bump
speed (z)
time (s)
over the session the bump advances 0.69 cm per cm run while the state holds, 0.38 once it is lost
the unit cell across the switch
122
128
134
140
146
152
158
164
170
six trials each, labelled by first trial
F
both states above the shuffle
phase shuffled
non-anchored
anchored
density
bump population vector, per 100 ms
G
24 sessions, 440 grid cells
both above zero: p = 1.2e-7
anchored higher: p = 1.3e-4
speeds equated: p = 7.6e-5
bump above shuffle
anchored
non-anchored
H
150 switches: it does not
decay into the new state
no step: p = 0.197
no decay after: p = 0.964
bump above shuffle
trial from the switch